

How to Write a Finance Research Proposal

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What Makes a Strong Research Proposal?

A research proposal must clearly address:

- **Research question:** specific and testable
- **Motivation:** real-world and academic relevance
- **Hypothesis / mechanism:** direction and economic intuition
- **Data:** source, sample, and time period
- **Methodology:** event study, panel regression, DiD, etc.
- **Limitations:** identification concerns and risks

Sharpening a Question: Bad → Good

Too vague:

“I will study the effects of macroeconomic factors on stock performance.”

What is missing? Outcome (Y), driver (X), sample, timeframe, method.

Sharpened, step by step:

- “macroeconomic factors” → *inflation* → *inflation surprises around CPI release days*
- “stock performance” → *DAX banks* → *2015–2024 daily abnormal returns*

Result: “Do inflation surprises on CPI release days move DAX bank stock returns, 2015–2024?”

From Idea to Proposal: ESG Controversies

Q: “Do ESG controversies increase firms’ cost of equity?”

Motivation. ESG ratings increasingly steer capital allocation, yet the effect of *controversies* (vs. ratings) on the cost of equity is under-documented in Europe.

Hypotheses.

- H1** Controversies raise the cost of equity. (*main*)
- H2** Larger for firms with weaker pre-controversy ESG. (*extension*)

Design at a glance

Sample	German listed firms, 2016–2024
Sources	LSEG ESG; Yahoo Finance; Compustat Global
Y	Cost of equity proxy: Earnings yield (E/P) (or CAPM R_E)
X	Controversy $\times$ After (treated vs not, before/after)
Controls	firm size, leverage, industry
Method	Difference-in-differences with industry & year FE (or firm & year FE)
Limitation	treated & control firms must look similar <i>before</i> the event

From Idea to Proposal: Green vs Conventional Bonds

Q: “Are green bonds priced differently than comparable conventional bonds?”

Motivation. A negative “greenium” would mean investors accept lower yields for the green label — relevant to issuers’ funding costs and the credibility of green finance.

Hypotheses.

H1 Green bonds trade at lower yields.
(*main*)

H2 Greenium is larger for investment-grade issuers.
(*extension*)

Design at a glance

Sample	Green + matched conventional bonds, 2015–2024
Sources	LSEG Workspace/Eikon; Compustat
γ	Yield spread over government benchmark
χ	Green-bond indicator
Controls	maturity, rating, currency, coupon, size, sector
Method	Cross-sectional OLS regressions, controlling for bond features, time and industry FE
Limitation	green bonds are still a young, small market — skewed to recent years and large issuers

Common Mistakes (and How to Fix Them)

Too broad: “Does inflation affect the stock market?”

Fix: Narrow market, period, outcome, and design.

No measurable variables

Fix: Define Y , main X , *controls*, and unit of observation.

Correlation as causation

Fix: Acknowledge the limitations. Use a before/after comparison around an event (event study, simple DiD).

Too many hypotheses

Fix: One main hypothesis; one extension and no more.

Data not accessible

Fix: Confirm the source that delivers Y *before* committing

Literature = summary only

Fix: Position the paper: what we know vs. what you add.

Using AI Responsibly

AI can help you work faster — but you are in charge.

Appropriate uses:

- Editing: improve clarity, flow, and writing style
- Creativity: generate alternative research questions and hypotheses
- Design: suggest variables, data sources, and methods
- Reviewing: identify weaknesses in arguments / methodologies
- Iteration: draft → critique → revise

Using AI: Prompt Patterns (1/3)

Three ingredients: *constrain* the output, feed it *your own* work, then *verify*.

Vague → **constrained** (same idea, but now answerable):

- **Weak:** “Tell me about ESG and stock returns.”
- **Better:** “Give me 5 *testable* questions linking ESG controversies to cost of equity; rank by data feasibility and say why.”

What made it better? It *limits* the answer (5 questions, ranked) and points at a *specific* topic — so the AI can't wander.

Using AI: Prompt Patterns (2/3)

More pattern examples — the *constraint* is in brackets:

- **[your work + focus]** “Here is my question, Y , X , and method *[paste]*. List the 3 biggest threats to identification and one fix for each.”
- **[word limit + audience]** “Rewrite this motivation in ≤ 120 words, clearer and finance-focused *[paste]*.”
- **[paste + dimensions]** “Compare these two abstracts *[paste A and B]* on question, data, identification, and contribution.”

Notice: each good prompt either *limits* the answer (count, ranking, word cap) or *gives the AI your material* instead of asking it to invent facts.

Using AI: Prompt Patterns (3/3) — Empirical Work

For Stata / R/..., same rules, some examples:

- **[tool + task]** “In Stata, regress |return| on a treatment×after dummy plus size and leverage; explain each line.”
- **[paste error + fix]** “Here is my error message and the code that caused it *[paste]*. What is wrong and how do I fix it?”
- **[paste + check]** “Check this code against my design: are Y , X , and the sample what I described? *[paste]*”
- **[clean]** “This data has duplicate firm-year rows and missing returns *[paste a few rows]*. Drop duplicates and flag missing values. Give me the code in R.”
- **[merge / match]** “Merge my returns and ESG files on firm ID and date; report how many rows match.” / “Match each green bond to a comparable conventional bond (issuer, currency, rating, closest maturity). Give me the code in Python”

Notice: Make sure that you are allowed to paste/upload the data you are using.

Using AI: Verify everything

- ☐ **Citations:** the paper really exists — find the DOI/journal; authors and year are correct.
- ☐ **Numbers & dates:** cross-checked against the original source (ECB site, filing, database).
- ☐ **Variables:** the dataset actually contains them, defined as you assume.
- ☐ **Method matches your design:** right outcome, regressor, sample and controls — and any extras (e.g. fixed effects, clustered SEs).
- ☐ **Code runs:** you ran it, no errors, and the output makes sense (signs, sizes, no. of obs).
- ☐ **Understanding:** you can explain every line and claim *without* the AI.

Rule: If you cannot explain or defend it, do not use it.

Academic integrity — the work must be yours:

- AI is a tool, not a co-author; the ideas, analysis, and writing are your responsibility.
- **Disclose** your AI use (which tool, and what for); “the AI said so” is not a defence.
- Never submit fabricated citations or results; follow the university’s policy.